

Introduction

In 2023, the NFL generated \$20 billion in revenue, averaging 17.9M viewers per game, an indication of its importance to teams and fans, as well as analysts and coaches who often use tracking data to understand quarterback movements and decision making. With quarterback evaluation and development so highly sought after, we created a model that uses NFL games data to determine the optimal receiver for QBs to pass to at a given moment and compared it to the results of the actual play. We also incorporated this model into an interactive interface for the use of analysts and players alike.

Data

The NFL Big Data Bowl 2025 dataset from Kaggle provides comprehensive football play data across multiple dimensions. The dataset contains five main files. These include game data (teams, scores, dates), play data (detailed play-by-play information including formations, pass results, and expected points), player data (physical attributes and demographics), player-play data (individual player statistics and actions for each play), and tracking data (player movement coordinates, speed, and orientation).

We were able to take these 5 files and manipulate them by creating new features based on the goal of our project. We were able to process the play data by engineering completion rates, pass characteristics, and game situation metrics. We also were able to capture target locations, pocket time, and throw timing. We also processed the player data and converted heights to inches and calculated BMI, created size-based receiver classifications (small/medium/big) and developed QB experience categories (rookie/young, prime, veteran, senior). From the tracking data, we extracted movement metrics (speed, acceleration, direction) and created play phase distinctions (pre-snap, post-snap, post-pass).

identifiers	quarterback features	qb decision features	multi-week player tracking features
gameId	displayName	completion	displayName
playId	height_inches	epa	s_mean
nfId	weight	epa_success	s_std
	age	yards_gained	a_mean
	experience_level	downs	a_std
	bmi	yards_to_go	dis_sum
		yards_from_goal	movement_alignment_mean
		red_zone	x_min
		expected_points	x_max
		end_of_half	y_min
		pass_length	y_max
		pass_type (5)	week_first
player features	receiver target features	target_location (3)	x_distance
displayName	was_targeted	time_to_throw	y_distance
position	route_ran	time_in_pocket	route_directness
position_group	coverage_assignment	pressure	direction_changes
height_inches	down	play_action	dis_sum (5)
weight	yards_to_go	dropback_type	frameId_count (5)
age	yards_from_goal	shotgun	s_max (5)
is_receiver	quarter	empty	s_mean (5)
receiver_type		third_and_long	
bmi		goal_to_go	
height_vs_db		deep_in_own_territory	
weight_vs_db		red_zone_pass	

Figure 1: Preprocessed Data

Methodology

With the data, we created two different models. The first model utilizes a random forest architecture to assess how quarterbacks currently make decisions. We use our engineered features to predict which receiver a quarterback is most likely to target. We prepare the training data by looking at receiver's data on passing plays so our model can learn patterns on quarterback decision making. The target variable was a binary classifier under the was_targeted column indicative of whether the receiver was the intended target or not.

We performed an 80/20 train-test split and ran a grid search cross validation on the number of estimators (100) and the maximum tree depth (20) to identify the best hyperparameters. To ensure good generalization of our model, we set the maximum number of features to 'sqrt'. Our model outputs probability scores for target recommendations mapped to four categories being a strong target, good option, secondary option or low priority.

The second model utilizes data regarding the position, velocity, and role of each player on the field. We explicitly encode information that allows the model to learn whether the player is on offense or defense, whether they were rushing the passer, whether they were a route runner, and whether they were the receiver actually targeted, to predict three outputs: (1) the probability of a completion, (2) the probability of an interception, and (3) the expected yards gained if the pass is completed. In practice, we treat each pass attempt as having three possible outcomes - completion, incompletion, or interception - with which we can then calculate the Expected Points Added (EPA) for.

We tracked target percentage, average EPA, completion percentage, interception percentage, mean yards gained, and mean completion yards gained for each ranking from 1 to 5, in plays where the quarterback targeted that receiver and used the frame half a second prior to the pass attempt to account for the time it takes for a quarterback to pass a ball after having made their decision.

```
=== Red zone play ===
Recommendations for Game 2022090800, Play 56:
  displayName  recommendation  target_probability  route_directness  was_targeted
Stefon Diggs   STRONG TARGET      0.711974           0.556191           1
Isaiah McKenzie STRONG TARGET      0.527305           0.772786           0
Dawson Knox    SECONDARY OPTION    0.274297           0.738127           0
Gabe Davis     LOW PRIORITY        0.000101           0.824823           0
✓ Model recommendation matches the actual target!
```

Figure 2: Sample Output from Random Forest Model

Results

The random forest classification model was successfully able to incorporate player and play-level features to gain insight on quarterback decision making, with an accuracy of 86.69% in our model without multi-week tracking data, and 81.72% with it. We found that the most influential feature in a quarterback's on field decision making is pass length, with a feature importance of 11.01% in models trained without multi-week data and 13.55% with it. Additionally, target location metrics such as width, depth and distance were consistently important across models, with feature importances up to 8.86%, 6.14%, and 6.10% respectively. Although players' physical attributes had minimal overall impact on model performance, our model suggests that a player's weight matters more than height.

The transformer-based model was successfully able to utilize position, velocity, and role information of each player on the field to recommend a receiver that would provide the greatest increase in EPA. We found that when quarterbacks passed to higher ranked receivers per our model, they observe a considerably higher average EPA, confirming that quarterbacks would be more effective when following our model recommendation.

In order to visualize the true three-dimensional nature of the sport, we have created an application capable of displaying all player locations and their movements for any play in our data set. To use the application, the user is introduced to a dropdown menu where they would have to select the game and then the play they would like to analyze. After selecting the play, a visualization is created displaying the players as cylinders and the ball as a brown sphere. During the play, the top three modeled receiving options in a given frame are colored gold, silver, and bronze respectively. Each play also includes a line chart that demonstrates how each receiver's modeled expected EPA evolves throughout the play (Figure 2). This platform will enable quarterbacks and coaching staff to visually interpret model-suggested targets in real-time scenarios.



Figure 3: Visualization with Transformer Model

Conclusions

Our findings suggest that static player attributes—such as position group, height, and experience level—as well as defensive coverage assignments, are less significant indicators of quarterback decision-making than dynamic, context-driven features. Features related to location and movement, including x-position, speed, route type, and distance from the goal line, appear to be more influential, which aligns with our findings that quarterbacks tend to favor receivers positioned in areas associated with higher completion probabilities, even if those areas do not always yield the highest EPA.

By integrating these analytical insights into a 3D visualization tool for play-by-play review, we offer a platform that enables quarterbacks and coaching staff to visually interpret model-suggested targets in real-time scenarios. This enhances decision-making by providing a clearer understanding of optimal passing options grounded in data rather than intuition and experience alone.

Our work also highlights the importance of temporal and situational factors—such as time to throw, game quarter, and proximity to the end zone—in influencing throw decisions. In future iterations, incorporating more granular situational context (e.g., down, score differential, or pressure data) into our transformer-based recommendation model could further improve its utility and realism. This would allow for even more nuanced, game-aware suggestions that adapt to high-pressure or time-critical situations.